

Scope-completion strategy toward A1-relative unconditional Reading-H T7:

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope: The operator decomposition (accepted) and the question

The enacted T7 is unconditional in the comparison hypothesis H-LAYER but rests on three standing inputs: (1) A1-KERNEL-CONV (the microscopic-theory kernel-convention *definition*, T5); (2) the lattice competitor class $\mathcal{C}_{\text{phys}}$ (real antipodal crystallographic, packing-bounded, coherence-resolved); (3) the operating window $I = 2 \times 10^{-3}$, $\mu^2 \in [\times 0.5, \times 2]$. The operator correctly observes that an *absolute* assumption-free T7 is not meaningful — every theorem about F_{TECT} is relative to the definition of F_{TECT} — and proposes the realistic target

$$\mathbf{A1\text{-}relative\ unconditional\ Reading\text{-}H\ T7:} \quad F[Q] - F[G_*] > 0 \quad \forall Q \in \mathcal{C}_{\text{full}} \setminus \{G_*\},$$

$\mathcal{C}_{\text{full}}$ widened to all admissible competitors (lattice and non-lattice), over a wider window, keeping only A1. The question: how hard, and by what route?

2. Ruthless check of the “raise DR-2 to theorem-grade” framing

The operator suggests non-lattice closure “probably needs DR-2 fallback raised to theorem-grade”. This conflates two different roles of the lattice restriction. The off-diagonal stability is governed by the single N -independent ratio

$$R_{\text{lead}}(Q) = \text{const} (1 + K_{\text{floor}}(Q)) I, \quad \text{const} = \frac{9}{4} u_{\text{eff}}^2 B_{\text{max}} \frac{2}{r_R} = 23.2,$$

$$R_{\text{lead}} < 1 \iff K_{\text{floor}} < K_* = \frac{1}{\text{const} I} - 1 \simeq 20.5,$$

where $K_{\text{floor}}(Q) = \sum_{t \neq 0} (\text{sum-circle occupancy})^2 / N^2 - 1$ is the additive-energy floor. DR-2 ($E_+ \leq_\epsilon N^{2+\epsilon}$) bounds K_{floor} in the *worst* (arithmetic) case. The probe res5_031 measures K_{floor} directly:

Configuration (non-lattice unless noted)	K_{floor}	R_{lead}
random real-antipodal, $N = 12-60$	0.75–0.95	~ 0.09 (margin $\times 11$)
coplanar regular ring (max symmetry), $N = 8-40$	0.62–0.93	~ 0.08
jittered-BCC (10% off-lattice)	0.92	~ 0.09
<i>exact</i> BCC (lattice)	1.75	–
admissible lattice class (pinned)	≤ 13	≤ 0.65
threshold K_*	20.5	1

Every non-lattice configuration tested has $K_{\text{floor}} \sim 1 \ll 13 \ll 20.5$, and perturbing a config OFF the exact lattice REDUCES K_{floor} . The additive-energy floor is maximised by exact arithmetic coincidence of sum-circles; a non-lattice config has generic (non-degenerate) sum-circles, occupancy ~ 2 , hence $K_{\text{floor}} \sim 1$.

Conclusion (reframing). The lattice class is the off-diagonal *adversarial worst case*, not an easy subcase. The enacted lattice T7 already covers the hardest competitors of the R_{lead} mechanism; generic non-lattice competitors are $\times 11$ more stable. Re-opening arbitrary- Q DR-2 is the wrong target — it bounds the lattice/worst case, which is *already* T7 for the lattice.

3. The genuine residual: an additive-energy extremal conjecture (T2)

What “ $\mathcal{C}_{\text{full}}$ ” actually requires is the rigidity/extremal statement

(EXT) $\max_{Q \in \mathcal{C}_{\text{full}}, |Q|=N} K_{\text{floor}}(Q)$ is attained within the arithmetic (lattice) sub-class, and is ≤ 13 .

(EXT) is the honest bridge from “lattice T7” to “ $\mathcal{C}_{\text{full}}$ T7”: if the additive-energy floor is maximised by arithmetic structure (a Freiman/Cauchy–Davenport-type extremal principle on S^2), then no admissible non-lattice competitor exceeds the lattice bound and the lattice T7 dominates all of $\mathcal{C}_{\text{full}}$. The probe is STRONG EVIDENCE for (EXT); its status is T2 CONJECTURE with the pre-registered falsifier

Falsifier: a real-antipodal admissible Q with $K_{\text{floor}}(Q) > 13$.

This is a tractable additive-combinatorics statement, not the (harder, partly OPEN) arbitrary- Q ℓ^2 -decoupling programme.

4. The (I, μ^2) window is bounded by physics, not by a proof gap

Intensity. R-029 (SC-SCOPE) records a physical critical intensity $I_c \sim 2.5 \times 10^{-3}$ *beyond* the operating endpoint, where the all-orders joint flips sign. The honest maximal intensity window is therefore $I \in (0, I_c)$ with $I_c \sim 2.5 \times 10^{-3}$ — a phase boundary of the selection, NOT a gap to be closed. Claiming $I \in (0, \infty)$ would be false. **Mass.** $\mu^2 \in [\times 0.5, \times 2]$ was margin-checked (SC-SCOPE worst $\times 1.034$); widening it is a tractable scan plus a monotonicity argument, with the thin selection margin 0.097 as the risk to monitor.

5. The status of A1-KERNEL-CONV (the floor, with a standing obligation)

A1 is a *definition*, so “A1-relative unconditional” is the correct floor: one cannot remove the kernel convention without removing the theory. One honesty caveat must not be glossed: A1 is T5 (*pinned*, not *derived*). For the comparison theorem this is the right floor; for the larger TECT/TOE ambition, “why this kernel?” remains a standing Sector-A obligation (promote A1 $T5 \rightarrow T7$ = derive the kernel from the topological/condensate structure rather than fixing it by convention). “It is just a definition” is correct for the present scope and incomplete for the full-theory claim.

6. Recommended path (honest tiers)

1. **Cheapest real gain — pin the window (days).** Set I_c rigorously from the SC-SCOPE joint (replace “ $I = 2 \times 10^{-3}$ endpoint” by “ $I \in (0, I_c)$, I_c certified”); scan μ^2 outward with a monotonicity certificate. Removes one scope qualifier honestly; no new mathematics.
2. **The real theorem — prove (EXT) (weeks).** Promote the extremal conjecture $T2 \rightarrow T6/T7$: “the S^2 additive-energy floor is maximised by arithmetic structure”. This is the single statement that upgrades “lattice T7” to “ $\mathcal{C}_{\text{full}}$ T7” for the off-diagonal mechanism. Re-use R-025/R-026 (sum-circle richness, lattice DR-2) as the arithmetic-side input.
3. **Complete the decomposition for non-lattice — (D) and (S) (weeks).** The probe covers the off-diagonal (O) only; confirm the diagonal isotropy (D, T-016) and the selection floor (S) extend to non-lattice (expected easier: less structure, lower energy).
4. **External referee (standing).** The publication-grade dual audit.
5. **A1-relative consolidation.** Once 1–3 close, restate the theorem on $\mathcal{C}_{\text{full}}$, $I \in (0, I_c)$, given A1.

7. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “The probe is a finite sample; it cannot prove (EXT).” CORRECT and stated: (EXT) is graded T2 CONJECTURE, not a theorem; the probe is STRONG EVIDENCE with a pre-registered falsifier ($K_{\text{floor}} > 13$ for an admissible non-lattice Q). No tier change is claimed.
- (β) “Generic non-lattice is easy, but an exotic high-energy non-lattice config might beat the lattice.” This is exactly what (EXT) must exclude; the probe tests random + maximal-symmetry + near-lattice and finds none. The arithmetic-maximises-energy heuristic (lattices maximise additive energy) is the expected mechanism; proving it is item 2.
- (γ) “Does this widen the enacted T7 now?” No. B1/B2 stay at the enacted scope-qualified T7; this note is a roadmap + a T2 conjecture, not a promotion.

Quantitative sanity check (mandatory): *threshold* $K_* = 1/(23.2 \cdot 2 \times 10^{-3}) - 1 = 20.55$; *worst non-lattice* $K_{\text{floor}} = 0.95 < 13 < 20.55$; *lattice-is-worst* exact-BCC $1.75 > \text{jittered } 0.92$; *window* $I_c \sim 2.5 \times 10^{-3}$ (R-029). All from res5_031 (5/5).

8. Result footer

Result ID:	B1/B2 scope-completion strategy toward A1-relative unconditional T7
Precise statement:	The lattice class is the OFF-DIAGONAL adversarial worst case (R_lead governed by N-indep const=23.2, $K_*=20.5$; every non-lattice config $K_{\text{floor}} \sim 1 < 13 < 20.5$; off-lattice perturbation reduces K_{floor}), so the enacted lattice T7 dominates non-lattice competitors of the R_lead mechanism. Residual to C_full = additive-energy EXTREMAL conjecture (EXT) “ K_{floor} maximised by arithmetic structure” (T2, strong evidence; falsifier $K_{\text{floor}} > 13$), NOT arbitrary-Q DR-2. The (I, μ^2) window is bounded by a PHYSICAL critical $I_c \sim 2.5e-3$ (R-029), not a proof gap. A1 is the definitional floor (standing derive-the- kernel obligation for the full theory).
Scope:	Off-diagonal mechanism (probed); (D)/(S) non-lattice extension not probed. No tier change. B1/B2 stay enacted scope-qualified T7.
Dependencies:	t7-enactment-record v1.0; R-025/R-026 (sum-circle/lattice DR-2 T7); R-029 (SC-SCOPE critical I_c); T-016 (D), T-018 (R_lead/ K_* identity).
Evidence grade:	STRATEGY + T2 CONJECTURE (EXT) + EXECUTED (res5_031_nonlattice_extremal_probe.py 5/5).
Reproduction command:	python codes/vacuum/res5_031_nonlattice_extremal_probe.py
Expected output:	$K_*=20.55$; non-lattice $K_{\text{floor}} 0.6-0.95 < 13$; exact-BCC $1.75 > \text{jittered } 0.92$; 5/5 PASS.
Falsifier:	A real-antipodal admissible non-lattice Q with $K_{\text{floor}} > 13$ (refutes

EXT); or a selection margin going negative
inside $I < I_c$, μ^2 widened.

Tier before / after: No change. B1/B2 enacted scope-qualified T7.
(EXT) registered T2.

No-overclaim statement: Reframing + roadmap; NOT a widening of the
enacted T7. (EXT) is a
conjecture with strong evidence, not a theorem.
Window bounded by
physics. A1 remains the definitional floor.

Next required action: (1) pin I_c + scan μ^2 (window); (2) prove
(EXT) $T2 \rightarrow T6/T7$
(the real theorem); (3) extend (D)/(S) to
non-lattice; (4) referee.